

Software Engineering Take-Home

ARP Digital – Global Payments Service

1 Background: How ARP GPS Works

ARP Digital makes money by facilitating remittance payments out of the UAE and Bahrain with low fees. For each payment, we convert the sender's funds into USDT in the UAE, and our partner in the destination country converts that USDT into local currency. Our margin comes from the rate difference between USDT and USD in the destination country: in many emerging markets USDT trades at a premium to the US dollar, and we capture that premium on every transaction.

To grow the business, we need to identify destinations where two conditions hold: meaningful remittance volume from our source markets, and a sizable USDT premium. This assignment is about visualizing both.

2 The Task

Build an interactive world-map dashboard that helps ARP's strategy team identify the most attractive new corridors to expand into, from the UAE and from Bahrain.

You will combine three data sources into a single visualization, ranked by a *composite score* that captures both the size of the remittance flow and the size of the USDT premium in each destination. Data can be fetched by a one-time script – no caching or scheduled refresh is required.

3 Data Sources

3.1 World Bank KNOMAD Bilateral Remittance Data

Use the attached file `WB-KNOMAD.xlsx`.

- **Sheet:** Data
- **Indicator:** `WB.KNOMAD.BRE` – “Bilateral Remittance Estimates using Migrant Stocks, Host Country Incomes, and Origin Country Incomes (US\$ million)”
- **Sender filter:** `Economy Name` $\in$ {United Arab Emirates, Bahrain}. In this dataset, `Economy` is the *sender* and `Partner` is the *destination*.
- **Year:** 2021 is the most recent year with populated UAE and Bahrain data. Use the value in the 2021 column.
- **Exclude:** rows where `Partner` = "WORLD" (these are aggregates, not corridors).

Each remaining row represents one outbound corridor, with the 2021 column giving the estimated annual flow in USD millions. You should find approximately 28 corridors per sender after excluding the WORLD row. This is not a filtering error – KNOMAD's model only produces estimates for country pairs with a measurable migrant diaspora, and these 28 corridors account for essentially 100% of each sender's total outbound remittance.

3.2 USDT / Local Currency Price – Binance P2P Express

For each destination currency, query the Binance P2P Express `quoted-price` endpoint to retrieve the reference USDT price in local currency. This endpoint is *unauthenticated* – no API key, no account required.

```
POST https://p2p.binance.com/bapi/c2c/v2/public/c2c/adv/quoted-price
Content-Type: application/json

{
  "assets": ["USDT"],
  "fiatCurrency": "<LOCAL_CURRENCY_CODE>",
  "tradeType": "SELL",
  "fromUserRole": "USER"
}
```

The response contains a `referencePrice` field under `data[0]` – this is Binance’s aggregate reference rate for the pair and is the number Express displays as the headline price. Use this directly; no pagination, averaging, or ad-ranking logic is required.

`tradeType: "SELL"` gives the reference price for selling USDT into local currency – i.e., the price *we* would receive when our payments partner converts USDT in the destination country.

Expected edge cases:

- Binance has withdrawn P2P from some jurisdictions. An empty `data` array or a null `referencePrice` is a valid outcome and must be handled gracefully – do not treat it as an error.
- Not every destination country maps cleanly to an ISO-4217 currency code. Maintain your own country-to-currency mapping (e.g., Egypt → EGP, Jordan → JOD, Sri Lanka → LKR). Document any omissions.

3.3 Official USD / Local Currency Rate – Oanda

Use the Oanda Exchange Rates API for the official USD-to-local-currency rate, or an equivalent free source such as Frankfurter or exchangerate-API.

3.4 Computing the Premium

For each destination currency c :

$$p_c = \frac{\text{USDT price in local currency on Binance P2P}_c}{\text{Official USD price in local currency on Oanda}_c} - 1$$

Expressed as a decimal (e.g., 0.143 for 14.3%). This is the fraction of extra value captured by selling USDT at the parallel rate versus the official rate – the margin per unit of flow routed through the ARP rail.

4 Dashboard Requirements

4.1 World Map

Render destination countries on an interactive world map as a heatmap colored by the currently selected metric. Countries that are not remittance destinations from either sender should appear in a neutral color.

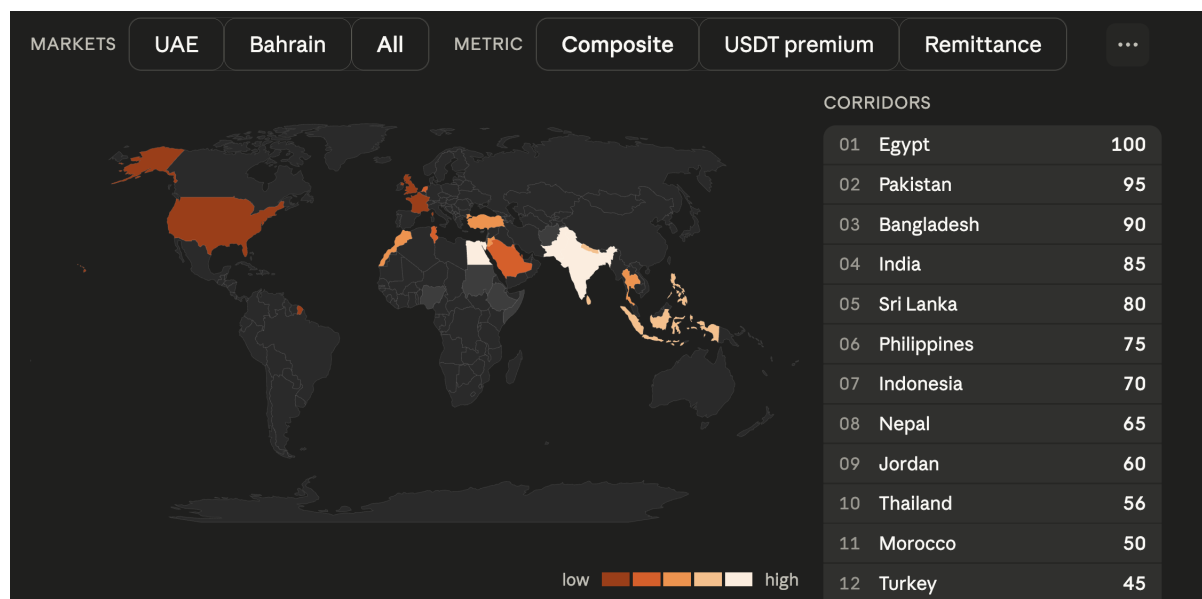


Figure 1: Reference dashboard mockup

4.2 “Markets” Toggle

A segmented control (buttons or tabs) that switches between:

- **UAE** – show only corridors where the sender is the United Arab Emirates.
- **Bahrain** – show only corridors where the sender is Bahrain.
- **All** – sum the remittance flows from both senders for each destination. USDT premium is a property of the *destination* (not the sender), so it does not need to be combined.

4.3 Metric Toggle

A second control switches the map’s heatmap coloring and the leaderboard ordering between three metrics:

1. **Composite Score** – integer 1–100 (see §5).
2. **USDT Premium** – percentage, one decimal place (e.g., 14.3%).
3. **Remittance Outflow** – USD, abbreviated (e.g., \$1.8B or \$1,833M).

Each metric has its own color scale and sort order.

4.4 Display of Numbers

The world map shows country fill intensity based on the selected metric. Numbers are **not** displayed on the map itself.

A leaderboard on the right-hand side of the dashboard lists all corridors sorted by the currently selected metric, showing the destination country name and the metric value.

Format numbers according to the selected metric:

- Composite Score: integer 1–100, e.g., 87
- USDT Premium: percentage with one decimal, e.g., 14.3%
- Remittance Outflow: USD with magnitude suffix, e.g., \$1.8B

5 The Composite Score

Let V_c be the annual remittance outflow to destination c in USD (from KNOMAD 2021, converted from millions), and let p_c be the USDT premium for destination c as a decimal.

Compute the raw addressable margin:

$$m_c = V_c \times p_c$$

Rank all corridors in the currently selected Markets filter by m_c and convert to a percentile in the range $[1, 100]$. Recompute whenever the Markets filter changes.

Missing Data

If a destination's currency has no Binance P2P data, **exclude** that corridor from the ranking rather than imputing $p_c = 0$. Surface this in the UI (e.g., "No P2P data available") so the strategy team can distinguish a genuine data gap from a zero-margin corridor.

6 Deliverables

1. Source code in a git repository (public or private, shared with the interviewer).
2. A `README.md` containing:
 - Setup and run instructions for running locally
 - How you used AI to produce the codebase, ensuring consistency, safety, and adherence to best practices

7 Evaluation Criteria

- **Correctness.** Numbers shown match the raw data; the math is right.
- **Edge-case handling.** Empty P2P responses, missing currency mappings, currencies where Binance has withdrawn coverage.
- **Code quality.** Readable, modular, reasonable error handling; easy for the next engineer to pick up.
- **Frontend clarity.** Can a non-technical strategy lead use this dashboard without explanation?
- **Writeup.** Does the README explain trade-offs honestly, including where you took shortcuts?

8 Notes on Stack

Use JavaScript end-to-end: React for the frontend and Node.js for the backend. Map library is your choice — `react-simple-maps`, `d3-geo`, `Leaflet`, and `Mapbox GL JS` are all reasonable options. Pick whichever you are most productive in.

If you have any questions about the assignment, please email the interviewer before sinking time into an assumption.